

# Levels of Aggregation in Election Data

Which rung of the ladder answers which question

84-355 Democracy's Data

August 2026

Somewhere in Alaska a voter ranked the Republican first and the Democrat second. 6,663 others made the same pair of marks in the same order, and 10,771 made it the other way round. Both numbers sit in a public file. Neither can be recovered from the vote totals of the precinct that voter lives in, or of the county, or of the state — not approximately, not with a better method, not ever. Add up the paper and the two numbers are gone.

Every claim about an election is a claim at some level of aggregation. *"Two-thirds of Alaskans used only one ranking"* is about ballots. *"This precinct swung twelve points"* is about precincts. *"Rural counties moved right"* is about counties. *"Trump won 312 electoral votes"* is about states.

These are not four resolutions of one picture. **Each step up the ladder permanently destroys a kind of question and permanently creates a new way to be wrong.** A photograph you zoom out of can be zoomed back in. An aggregate cannot.

## 01 The secret ballot is the deliberate destruction of individual-level data

When American states adopted the Australian ballot in the late nineteenth century they were not solving a data-management problem badly. They were making it impossible to know how any particular person voted, on purpose, because the alternative was a country in which employers and party bosses could check.

So the finest-grained thing that exists for almost every American election is **the precinct** — a few hundred to a few thousand ballots, added up. Everything we claim to know about *who* votes how, including everything a court is told about racially polarized voting under the Voting Rights Act, is an inference made across that gap. **The gap is the point of the institution and the problem of the method.**

## 02 Nothing here was downloaded

This chapter is the one place in the book with no source of its own. Every input is a file already committed to another chapter. Alaska's cast vote record. The Georgia Secretary of State's precinct returns for 2020 and 2024, with the vote-method breakdown. County returns for every state in 2024. Presidential results by congressional district.

National totals back to the nineteenth century.

The Election Assistance Commission's count of how many precincts and jurisdictions exist. And the Houston County, Georgia primary in which race is a recorded field.

Those published files are opened directly and every number on this page is re-derived from them. The result is kept as 17 small tables beside itself. Nothing was fetched, and nothing was retyped from another chapter's prose.

That is not laziness, and it is not tidiness either. **No American institution publishes the same election at more than one level of aggregation.** No American institution runs elections. The rungs of this ladder are produced by different bodies, with different obligations and no relationship to one another.

A state publishes precincts because its own law says so. Counties reach the public through a private compiler who stapled fifty-one state files together. Congressional-district results are an academic side product. The only file in which a row is a single ballot exists because Alaska adopted ranked-choice voting and cannot count the election without publishing one.

A chapter about the relationship between levels therefore has to be assembled. Building it out of chapters that already document their own sources is the only way to keep that history attached. Every difficulty those files have is inherited here whole.

If a number on this page is wrong, it is wrong in the chapter it came from too. That is the property worth wanting.

Be exact about the depth of that inheritance, though, because it is more than one layer. Follow the county file backwards. It arrives from a sibling chapter, which took it from a private compiler, who took it from fifty-one state canvasses. Three sets of hands before it arrives, and this chapter is the fourth.

What sits in the folders next door is the whole course's range of acquisition, compressed into one place. A state's own archives pulled down county by county. A compiler's stitched-together national file. An academic reconstruction of results by district. A federal survey of election offices. A ranked-choice ballot export that exists only because Alaska cannot count without it.

Every one of those histories is stripped off in transit. A file that arrives from a folder next door carries no address and no note saying what it cost.

The bill arrives in two forms. Inherited defects do not merely persist. They stack. A page assembled out of every one of those files is standing on all of their judgment calls at once, and no sentence here can tell you which one it is leaning on hardest. And this is the only chapter in the book whose data can go stale while nobody touches it. Rebuild any sibling with a renamed column and the work here stops with an error, which is the good outcome. Rebuild one with the same columns and different numbers and it will not stop at all.

It is the right evidence for two reasons that no better-designed source would supply. One is that the bottom rung exists at all: without a jurisdiction that publishes ballots, the claim that aggregation destroys information could only be asserted, never shown. The other is that two of these files carry an **answer key**. Georgia reports its precinct returns broken out by vote method, and Houston County records each voter's race on the registration form. So the guess this chapter runs across the gap can be scored against the truth instead of merely defended. Almost no work on ecological inference has that.

The honest limit follows directly from how the ladder was built. **These are not seven views of one election.** The ballots are Alaska's 2024 House race. The precincts are

Georgia's presidential returns. The counties and states are national. The unit counts come from a federal survey of election offices.

Figure 1 is a picture of a structure, not a measurement of a single contest. Where the chapter needs one election observed at two rungs at once, which it does at the point where the method is tested, it says which one and stays inside it. The cleaning was correspondingly small. The cast vote record arrives already collapsed to distinct ranking sequences with a ballot count beside each, so no individual record travels here. Every other input arrives as a table somebody else already checked.

### 03 What arrives from next door, and what leaves again

A file that comes from a sibling chapter looks like this. It is Georgia's 2024 presidential returns by precinct, exactly as committed in `ga-precinct-returns/data/`, first three rows. The columns run down the page because twelve of them do not fit across it:

| Column as it arrives | Row 1   | Row 2   | Row 3   |
|----------------------|---------|---------|---------|
| county               | Appling | Appling | Appling |
| precinct             | 1B      | 1C      | 2       |
| registered           | (empty) | (empty) | (empty) |
| ballots_cast         | (empty) | (empty) | (empty) |
| Donald J. Trump      | 921     | 724     | 541     |
| Kamala D. Harris     | 108     | 67      | 782     |
| Chase Oliver         | 2       | 0       | 1       |
| Jill Stein           | 0       | 0       | 3       |
| Claudia De la Cruz   | 0       | 0       | 0       |
| Cornel West          | 0       | 0       | 0       |
| total                | 1031    | 791     | 1327    |
| dem_two_party_pct    | 10.5    | 8.47    | 59.11   |

Two things about it are worth stopping on. **The columns are people's names**, typed by 159 separate county tabulation systems. That is fine for Georgia in 2024 and useless the moment a second election is put beside it, because no other election has a column called `Kamala D. Harris`.

And **the file carries no history at all**. There is no address in it. No note about the block that made the 2020 half a manual download. No record that two of those six candidates were struck from Georgia's ballot. All of that is written down one folder away, and none of it travels.

Here are the same three precincts after they have been read:

| Level    | Unit       | Dem | Rep | Two party votes | Dem share |
|----------|------------|-----|-----|-----------------|-----------|
| precinct | Appling 1B | 108 | 921 | 1,029           | 10.50     |
| precinct | Appling 1C | 67  | 724 | 791             | 8.47      |

| Level    | Unit      | Dem | Rep | Two party votes | Dem share |
|----------|-----------|-----|-----|-----------------|-----------|
| precinct | Appling 2 | 782 | 541 | 1,323           | 59.11     |

Four columns of substance where there were twelve, and every one of them is now a *role* rather than a name: a level, a unit, the Democratic vote, the Republican vote. That is the whole trick of this chapter. Once a precinct in Georgia, a county in Nevada and a congressional district in Ohio all look like that, they can be stacked in one table and compared. That is exactly what the next section does, and exactly what the file that arrived could not support.

| Level                  | Rows arriving | Rows kept | A row is                    |
|------------------------|---------------|-----------|-----------------------------|
| precinct               | 2,701         | 2,684     | one polling place's returns |
| county                 | 159           | 159       | one county's returns        |
| congressional district | 14            | 14        | one seat in the U.S. House  |
| state                  | 1             | 1         | one state's returns         |

## Four decisions, and one of them removes rows

**A name becomes a party.** Kamala D. Harris becomes `dem` and Donald J. Trump becomes `rep`. Nothing in the file says which is which; a person has to know. It is the least technical step here and the one that makes everything downstream possible.

**Everyone else is dropped from the denominator** — the bottom number, the thing you divide by. The share reported is the *two-party* share. So Chase Oliver's and Jill Stein's votes count toward no denominator anywhere on this page, and neither do the votes cast for the two candidates Georgia struck from its ballot. That is the convention in this literature and it is still a choice, and it means the percentages here will not match a percentage computed out of `total`.

**The unit is the county and the precinct together.** Precinct names are what a county typed and they repeat across counties, so the key is built by pasting the two with a separator: `Appling|1B`. There is no code to use instead.

**17 precincts disappear.** Georgia reports 2,701 and this chapter's ladder has 2,684.

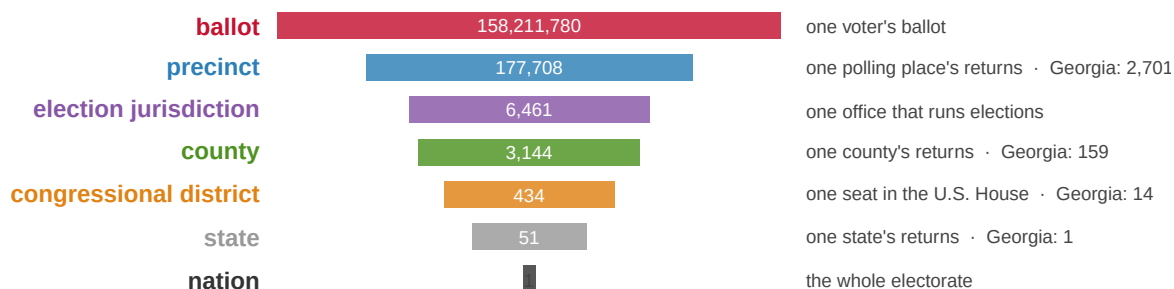
The missing ones cast no two-party votes at all. Their Democratic share is a division by zero rather than a small number, and a unit with no value cannot sit on a scatterplot of values. **Dropping them is right and it is not free.** Every count of precincts on this page counts precincts *where somebody voted for a major party*. That is not the same thing as a count of precincts. That difference is 17 places where people live.

## 04 Seven rungs, from a mark on paper to a number on the evening news

158,211,780 ballots were cast in 2024; 177,708 precincts counted them; 6,461 offices ran them; 3,144 counties, 434 districts and 51 states report them; one nation sums them.

**Eight orders of magnitude, and every step is one way.**

**every rung of the ladder, and how many units sit on it**



Each step up is irreversible. The rows below are not recoverable from the rows above.

**Figure 1. Every rung of the ladder, and how many units sit on it.** Bar width follows the logarithm of the count, so the 158,211,780 ballots and the one nation fit on the same axis. Georgia's count is given where the rung exists in Georgia. Each step up is irreversible, because the rows below cannot be rebuilt from the rows above.

## 05 The only American election data in which a row is a person

Alaska runs ranked-choice elections, and to run them it must publish a **cast vote record**: one row per ballot, carrying every ranking that ballot contained. Nothing else in American elections is filed at that grain.

| Quantity                                                | Value   |
|---------------------------------------------------------|---------|
| ballots in the file                                     | 340,778 |
| distinct ranking sequences                              | 926     |
| bullet-voting rate among ballots that ranked anyone (%) | 65.2%   |
| ballots that named a second choice                      | 115,127 |
| ballots that ranked the same candidate more than once   | 5,286   |

340,778 ballots collapse to only 926 distinct sequences, because 65.2% of the voters who ranked anyone ranked exactly one candidate and stopped. And 5,286 ballots name the same candidate in more than one slot — a thing that can only be seen, and only be counted, one ballot at a time.

It is also the only file in which the pair of numbers that opens this chapter can be looked up.

| Ranked first     | Ranked second    | Ballots |
|------------------|------------------|---------|
| Begich, Nick     | Howe, John Wayne | 24,386  |
| Begich, Nick     | Peltola, Mary S. | 6,663   |
| Peltola, Mary S. | Hafner, Eric     | 39,684  |
| Peltola, Mary S. | Howe, John Wayne | 13,890  |

What makes those counts unusual is not their size. It is their type. A crossing of first choice with second choice is a property of two marks on one piece of paper. **No total of any kind contains it.** A precinct that reports how many votes each candidate received has already thrown away which candidates appeared together on the same ballot. It has thrown it away in a way no later arithmetic undoes.

## 06 Two ballots in every thousand

The precinct is where American election data actually begins: Georgia publishes 2,701, the country about 177,708. Everything above was paid to get here. What it buys is **coverage**. The whole Alaska file, all 340,778 ballots, is 0.215% of the 158,211,780 ballots cast in the country in 2024 — two ballots in every thousand.

**There is no national cast-vote-record file, and there will not be one.** If you want to say something about the United States, you have to leave the bottom rung.

## 07 The mean does not move, and the spread collapses

Take one election, the 2024 presidential vote in Georgia, and read it at four levels. The two-party Democratic share at each:

| Level                  | Units | Mean dem % | Sd    | Min  | Max   |
|------------------------|-------|------------|-------|------|-------|
| precinct               | 2684  | 48.90      | 24.68 | 3.4  | 100.0 |
| county                 | 159   | 48.90      | 20.64 | 8.0  | 84.8  |
| congressional district | 14    | 49.78      | 18.45 | 31.2 | 86.4  |
| state                  | 1     | 48.89      | 0.00  | 48.9 | 48.9  |

**The mean does not move:** 48.90% across 2,684 precincts, 48.90% across 159 counties, 48.89% for the state. The district row is the odd one out at 49.78%. That file carries 14 percentages and no vote counts, so its mean is unweighted while every other rung's is weighted. The next section comes back to why.

**The spread collapses:** from a standard deviation of 24.7 points to 20.6. Georgia holds precincts at 3.4% Democratic and precincts at 100.0%. Its most and least Democratic *counties* are 8.0% and 84.8%. One picture holds both:

### the same election, read at four levels



**Figure 2. The same election, read at four levels.** One dot per unit, showing the two-party Democratic share of the 2024 presidential vote in Georgia at 2,684 precincts, 159 counties, 14 congressional districts and one state. The dashed line is the state total, 48.89%. Dot area follows the votes cast in the unit, above a floor that keeps the smallest units visible. The center of gravity never moves; the scatter around it disappears, from a standard deviation of 24.7 points across precincts to 20.6 across counties.

## 08 Where the arithmetic stops being clean

Two places the climb is not clean, and both matter.

| Quantity                  | Counted as                 | Against                            | Gap     |
|---------------------------|----------------------------|------------------------------------|---------|
| Georgia Democratic votes  | 2,701 precincts summed     | 159 counties, same source          | 0       |
| Georgia Democratic votes  | 159 counties, state source | 159 counties, national compilation | -11     |
| Georgia Republican votes  | 159 counties, state source | 159 counties, national compilation | -7      |
| National Democratic votes | 3160 counties summed       | certified national total           | -10,874 |
| National Republican votes | 3160 counties summed       | certified national total           | -7,781  |

**Summing all 2,701 precincts reproduces the county file exactly – zero difference.** But the *national* compilation of those same Georgia counties is 11 votes off. Summing all 3,160 county rows in the country lands 10,874 votes short of the certified national total, which is 0.0145% low. Small, not zero, and the reason is the second problem.

**The county rung does not exist in Alaska.** Check the national county file's keys against the Census Bureau's county list and 44 rows do not match. Alaska has no counties, so the file carries 37 **state house districts** instead (State House District 1, and so on). The District of Columbia contributes 7 **wards**. A "county-level analysis of the United States" is, for those rows, an analysis of something else.

**And one rung cannot be reached by adding at all.** Precinct returns record no congressional district, so district results are not summed. They are *rebuilt* by somebody who

assigned census blocks to districts and split precincts accordingly. That is an estimate wearing the clothes of a count.

It is worth the trouble, because the district rung answers something no other rung can.

**Georgia's median congressional district is 40.7% Democratic while the state is 48.9%.**

That is a gap of 8.2 points. Nationally it is 48.4% against 49.2%. That is a fact about how votes are packed into districts, and **a state total has no districts in it.**

One rung higher, the same thing happens. Trump took 49.8% of the national popular vote and 58.0% of the electoral college. The nation rung cannot see that either.

**Before you read on, write two numbers down.** Georgia publishes its precinct returns broken out by vote method. What follows deletes that breakdown, keeps only each precinct's two margins — *what share of its votes came by mail, what share went to Biden* — and runs the standard method used in voting-rights litigation to recover what share of mail voters chose Biden. **Write down your guess, and write down the largest value the answer could possibly take.**

## 09 Delete the answer, then try to recover it

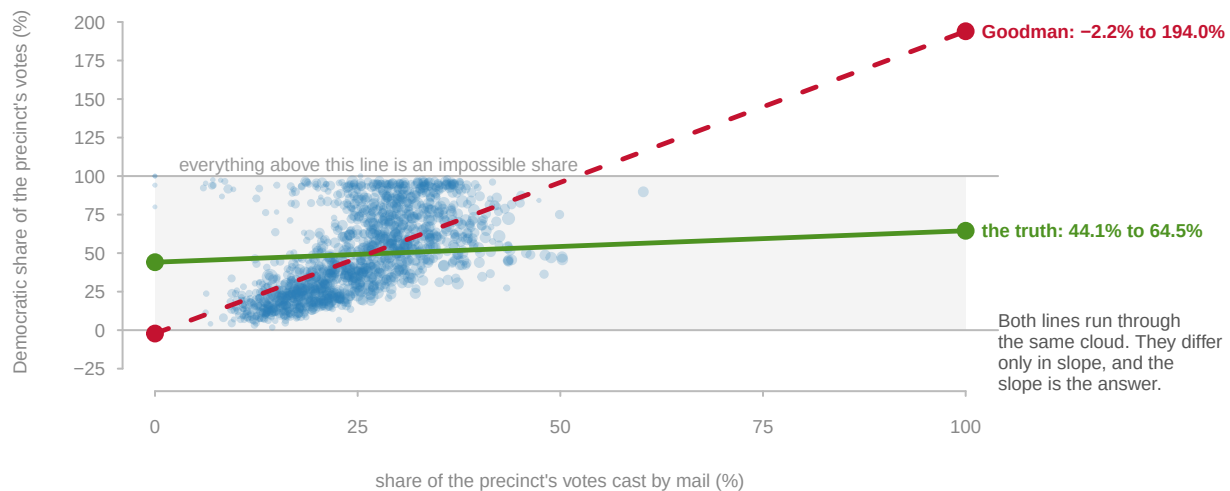
Georgia is unusual. It publishes the **joint distribution**, candidate by vote method, so the answer is not a guess at all. It is a table.

| Candidate       | By mail | Not by mail | Total     | % of them by mail |
|-----------------|---------|-------------|-----------|-------------------|
| Joseph R. Biden | 849,729 | 1,624,778   | 2,474,507 | 34.3              |
| Donald J. Trump | 451,157 | 2,010,680   | 2,461,837 | 18.3              |
| Jo Jorgensen    | 16,057  | 46,081      | 62,138    | 25.8              |

34.3% of Biden's Georgia votes came by mail against 18.3% of Trump's. Read the other way round: **64.5% of Georgia's mail votes went to Biden**, against 44.1% of everything else.

**Now delete that table.** Keep only what every other state publishes: for each precinct, the share of votes cast by mail and the share cast for Biden. Fit the line. This is Goodman's ecological regression, a best line drawn through a scatter of points and the workhorse of voting-rights litigation since 1953. and read off its value at 0% mail and at 100% mail.

### 2,653 Georgia precincts, 2020: mail share against Democratic share



**Figure 3. Two lines through the same cloud of precincts.** Each dot is one of Georgia's 2,653 precincts in 2020, placed by the share of its votes cast by mail and the share cast for Biden. Dot area follows the precinct's votes. The solid line joins the published truth at each end: 44.1% Democratic among votes not cast by mail, and 64.5% among votes cast by mail. The dashed line is Goodman's ecological regression fitted to the margins alone, read off at 0% mail and at 100% mail. The shaded band is the range a share can occupy; the dashed line leaves it.

The second number almost everyone writes down is 100, on the reasoning that a share of a group cannot exceed the group. That is the number the method breaks. **Goodman's regression, run on the precinct margins, says 194.0% of Georgia's mail voters chose Biden,** against a published truth of 64.5% — off by 129.5 points, and a share that cannot exist.

**Nothing in the aggregate data announces this.** The correlation is 0.60 and the line runs straight through the middle of the cloud — but so does the true line. **The two differ only in slope, and the slope is the entire answer.**

The reason is not a bug. Precincts with a lot of mail voting were Democratic precincts, and Democratic precincts were more Democratic *among mail voters and election-day voters alike*. The regression cannot tell "mail voters are Democrats" from "Democratic places mailed more," so it charges the whole between-precinct difference to the method of voting. **That is Robinson's ecological fallacy, in an election, with the answer key attached.**

## 10 Climb one more rung and it gets worse

Aggregate the same ballots from precincts into counties and run the same method:

| Level    | Units | R     | Goodman says (%) | The truth (%) | Bound, lower | Bound, upper |
|----------|-------|-------|------------------|---------------|--------------|--------------|
| precinct | 2653  | 0.599 | 194.0            | 64.5          | 15.2         | 98.4         |
| county   | 159   | 0.760 | 233.4            | 64.5          | 6.6          | 99.4         |

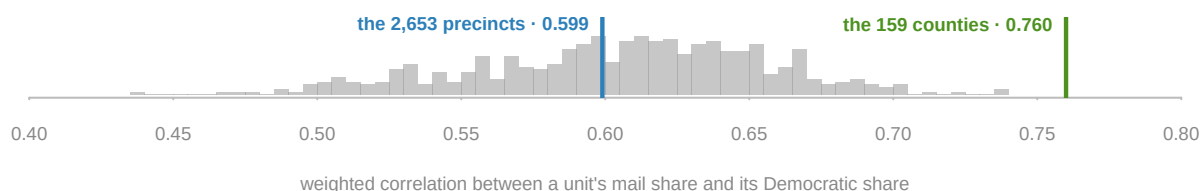
**233.4%.** The same ballots, grouped more coarsely, and the answer moves a further 39.4 points away from a truth that never changed.

The correlation rises too, from 0.599 to 0.760 – the standard result that **ecological correlations exceed individual ones**, which is why they are tempting. Is that a fact about voters, or about where Georgia drew its county lines? Regroup the same 2,653 precincts into 159 *random* groups, 500 times over:

| Grouping                                           | Units | Correlation | Note                                                         |
|----------------------------------------------------|-------|-------------|--------------------------------------------------------------|
| 2,653 precincts, ungrouped                         | 2653  | 0.599       | the finest unit with published returns                       |
| 159 real counties                                  | 159   | 0.760       | the same ballots, grouped by county line                     |
| 159 random groups of precincts (mean of 500 draws) | 159   | 0.605       | range 0.439 to 0.739; 0 of 500 draws reach the county figure |

**Random groups of the same size leave the correlation where it was** – 0.605 against 0.599 for the ungrouped precincts. Real counties push it to 0.760, which **none of the 500 regroupings reached**. So the lift is not produced by adding up as such. It is produced by adding up along *these particular lines*. That is Openshaw's modifiable areal unit problem: the statistic depends on the zoning, not only on the scale.

#### 500 random regroupings of the same precincts into 159 groups



**Figure 4. Five hundred random regroupings of the same precincts into 159 groups.** The histogram is the weighted correlation between a unit's mail share and its Democratic share under each random regrouping. The blue line is the ungrouped precincts, at 0.599; the green line is Georgia's real counties, at 0.760. Aggregating to groups of this size does not by itself lift the correlation. Aggregating along the county lines does, and none of the 500 random regroupings reached it.

**One thing the aggregate still supports honestly.** Duncan and Davis pointed out in 1953 that the margins alone rule some answers out with no model at all. Applied precinct by precinct and added up, they put Georgia's mail vote between 15.2% and 98.4% Democratic – wide, useless for litigation, and **correct**, which the regression was not. At county level they widen to 6.6%–99.4%.

## 11 The same failure, where it counts

Vote method is a convenient test case because Georgia published the answer key, but it is not what ecological inference is used for. The method is used to establish, in federal court, whether voters of different races prefer different candidates. There is an answer key for that too, in one place.

A Georgia county's voter file records each voter's reported race and which primary ballot they took, alongside the ordinary precinct margins. 4,293 of 4,631 Black voters took a Democratic ballot. That puts the truth at 92.7%. **Goodman's regression on the same precincts returns 102.1%.** Impossible again, and this time by 9.4 points rather than the

129.5 of the mail-vote case. Those are the precincts on which the first *Gingles* precondition gets litigated.

**The failure is not exotic.** It appears in a routine question about vote method, and in the central question of voting-rights litigation, for the same reason. The rung is above the level the question lives on.

## 12 What only the aggregate can answer

Everything so far argues downward. Four things argue back.

**Coverage.** A cast vote record exists for a rounding error of the 158,211,780 ballots cast in 2024. Any claim about the country is an aggregate claim or it is nothing. **Comparability.** A presidential series running from 1864 to the present exists at the state level and nowhere finer, because no smaller unit keeps its identity that long.

**The unit of the institution.** The electoral college is counted in states. A House seat is a district. A rejected ballot is reported by the jurisdiction that rejected it. When the institution adds up, the total is not a compromise. It is the object.

**Stability.** A precinct of four hundred voters is mostly noise. A county is not.

## 13 Choosing a rung

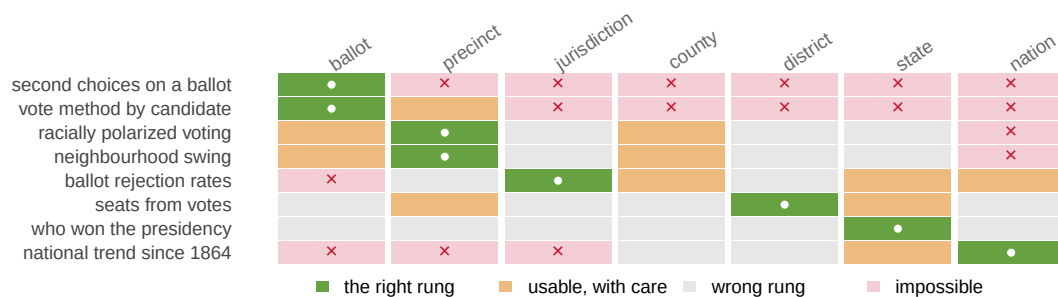
Everything above can be compressed into a single working rule: name the rung before naming the finding.

| The question                                                         | The rung                              | Why that one                                                             | The trap                                                                |
|----------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Did Begich voters who ranked a second choice prefer Peltola or Howe? | ballot                                | The two rankings must sit on the same ballot. No total can restore them. | Precinct returns for an RCV race give round-by-round totals, not pairs. |
| What share of a candidate's votes arrived by mail?                   | ballot, or precinct-by-method returns | It is a joint distribution. Georgia publishes it; most states do not.    | Regressing precinct Dem share on precinct mail share returns 194%.      |
| Is voting in this county racially polarized?                         | precinct                              | Smallest unit with returns; demographics attach through blocks.          | The estimate is an inference, not a measurement. Report the bounds.     |
| Which neighbourhoods swung between two elections?                    | precinct                              | County hides the variation; but check the boundary did not move.         | A precinct that 'swung' may just have gained three hundred houses.      |
| Which doors should a campaign knock on Saturday?                     | precinct                              | Fine enough to route a walk list, coarse enough to be stable.            | A precinct is not a person; targeting is a bet on an average.           |
| How many mail ballots were rejected, and for what reason?            | election jurisdiction                 | The jurisdiction is what administers and reports rejection.              | 'Jurisdiction' is a township in Wisconsin and a county in Texas.        |

| The question                                               | The rung               | Why that one                                                          | The trap                                                                 |
|------------------------------------------------------------|------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------|
| Did counties with less education move right?               | county                 | 'County' is the unit the education measure is published on.           | Robinson (1950): the county pattern need not hold for county residents.  |
| Does this map convert votes into seats fairly?             | congressional district | The seat is the district. Precincts must be aggregated into it first. | Districts do not nest into counties; someone had to disaggregate blocks. |
| Who won the presidency?                                    | state                  | The electoral college is counted in states, not in voters.            | The popular vote is not the quantity being decided.                      |
| Is the country's partisan balance shifting across decades? | nation                 | Only the top rung gives one comparable number over a long series.     | Anything below the top rung has changed shape underneath the series.     |

The same judgment, as a grid:

### which rung answers which question



**Figure 5. Which rung answers which question.** One row per question, one column per rung, shaded by whether that rung can answer that question at all. The green cells are the rung the question lives on. The red cells are questions that rung cannot answer, no matter how careful the analysis, because the information was destroyed on the way up.

The grid is the argument in one picture. Election data comes on a ladder, and every rung is a different object rather than a sharper view of one picture. The arithmetic of climbing is exact. 2,701 Georgia precincts summed to 159 counties reproduce the state total to the vote.

What the arithmetic hides is the price. From 2,684 precincts to 159 counties, 30.1% of the variation in Democratic share disappears. The 6,663 Alaskans of the opening page stopped existing a rung below that.

None of which licenses the opposite mistake. There is no rung that is correct in general; Openshaw's point is that the question is malformed. What can be done is narrower and more useful: name the rung, say why it is the rung the question lives on, and check whether the answer survives being asked one rung up. Goodman's estimate did not — it moved 39.4 points on the way from precincts to counties, on ballots that never changed.

## 14 Is aggregate-level inference legitimate at all?

The question is live, and not academic — it decides cases.

**The courts say yes.** In *Thornburg v. Gingles* (1986) the Supreme Court upheld a finding of racially polarized voting that rested on this kind of regression. Every §2 case since has run some version of the fit in Figure 3, on precinct margins.

**Freedman and colleagues said no, in exactly that setting.** Writing in 1991 about a suit against Los Angeles County, they argued that ecological regression depends on assumptions the data do not support. They offered a “neighborhood model” that fit the same data at least as well and gave different answers. That is the objection Figure 3 demonstrates. The method cannot tell “these people behave differently” from “these places are different.”

**King (1997) proposed a solution**, combining Duncan and Davis’s deterministic bounds with a model for where inside them the truth sits. It won prizes and it is the standard tool. It also drew sustained criticism, Freedman’s included, for substituting its own uncheckable assumptions and reporting intervals that are too narrow when they fail.

**The individual-level alternative raises the opposite objection.** Texas issued emergency guidance in June 2024 telling election officials to remove identifiers from released records. It warned that a combination of data points could link “a voter’s identity to their specific voting selections.” Courts elsewhere have ordered cast vote records released as public records. Both positions are defensible and they conflict.

**Notice where that leaves us.** Ecological inference exists because of the secret ballot; the secret ballot exists because the alternative was worse. **The methodological problem is a democratic achievement**, and the demand for individual-level data is a demand to partly undo it — an argument not against the demand but for knowing what is traded. The file underneath this chapter makes that trade itself: the Alaska ballots were collapsed from 340,778 rows to 926 distinct sequences, and the precinct and tabulator identifiers dropped, before anything was saved. **Even the bottom rung here is already an aggregate.**

## 15 What the ladder can testify to

The evidence is unusually clean in one respect. Every rung used above is a real published file rather than a simulation, and the walk between them is arithmetic that was checked rather than assumed. The pivot is stronger still, because Georgia publishes the joint distribution of candidate by vote method, which means the ecological estimate can be scored instead of argued about. Very little work in this area has an answer key at all.

It is also narrower than it looks, in four specific ways. **One state carries most of the walk.** Georgia publishes precinct returns by vote method and most states do not. Whether Goodman’s regression fails this badly elsewhere is not shown here and should not be assumed.

**One contest carries the bottom rung.** Alaska’s cast vote record exists because the election was ranked-choice. That is exactly why nothing here can say what such a record would reveal about an ordinary plurality race.

**The district rung is borrowed rather than built.** It arrives as percentages with no vote counts, which is why its mean and spread in Figure 2 are unweighted while every other rung’s are weighted. That is a difference in construction, not a finding about districts.

And **vote method is not a demographic**. “Mail voter” is a behaviour with no census count behind it. The racially polarized case, where the group is fixed and the stakes are a federal remedy, is harder than the one worked through here.

Three questions stay open, and none of them will be closed by a bigger version of this table. Whether cast vote records should be public at all: litigation is active and the states disagree, with privacy and verifiability both real and both on the line. Whether King's method solves the ecological inference problem or relocates it: thirty years of argument have not settled that. And how to report an ecological estimate honestly, given that the deterministic bounds are correct and nearly useless while the point estimate is useful and up to 129.5 points wrong. No accepted practice puts both of those in front of a court, which is the practical shape of the problem this chapter has been describing.

## 16 Sources

**Data.** Nothing here was downloaded. Every number above is re-derived from files other chapters of this book already fetched and committed, and each of those chapters documents the fetch. The publishers underneath are these.

- State of Alaska, Division of Elections, 2024 general election cast vote record — the bottom rung, one row per distinct ranking sequence. <https://www.elections.alaska.gov/results/24GENR/>
- Georgia Secretary of State, certified precinct returns for the 2020 and 2024 general elections, with the 2020 vote-method breakdown. <https://sos.ga.gov/page/historical-elections-results>
- Tony McGovern, *United States General Election Presidential Results by District, Ward, or County from 2008 to 2024* — county and state returns for the whole country. [https://github.com/tonmcg/US\\_County\\_Level\\_Election\\_Results\\_08-24](https://github.com/tonmcg/US_County_Level_Election_Results_08-24)
- Jason Timm, *PresElectionResults* — results by congressional district for 2024, and the national series back to 1864. <https://github.com/jaytimmm/PresElectionResults>
- U.S. Election Assistance Commission, *Election Administration and Voting Survey* — the national counts of each kind of unit. <https://www.eac.gov/research-and-data/datasets-codebooks-and-surveys>
- Houston County, Georgia, General Primary of 21 May 2024, with self-reported race — the graded ecological case. A public record in Georgia, obtained as working material rather than downloaded, and described where this book first uses it, in the RPV chapter.

Also cited: Texas Secretary of State, *Election Advisory No. 2024-20*, emergency guidance on voter privacy, 6 June 2024.

### The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`consistency.csv`, `cvr_facts.csv`, `cvr_second.csv`, `decision.csv`, `eco_county.csv`, `eco_estimates.csv`, `eco_precinct.csv`, `ga_levels.csv`, `ladder.csv`, `level_stats.csv`, `maup_draws.csv`, `maup_null.csv`, `method_truth.csv`, `not_counties.csv`, `rpv_check.csv`, `seat_vote.csv`, `suitability.csv`

### Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `suitability.csv` scores each question against each rung of the ladder. Pick a question you care about and find the rung that fits. Then check what the published data actually offers at that rung.
- `eco_estimates.csv` shows what an ecological estimate gets wrong, with `truth_by_mail` beside the estimate. Work out the size of the error. Would you have known it was wrong without the truth column?
- `maup_draws.csv` recomputes the same correlation on many random groupings. Look at the spread. How much of a published correlation is the grouping rather than the people?
- `ladder.csv` gives `share_of_voters_pct` and what is `published_for` at each rung. Find the rung where publication stops. What questions die there?

### Academic literature

- Robinson, W. S. (1950). "Ecological Correlations and the Behavior of Individuals." *American Sociological Review* 15(3): 351-357.
- Goodman, L. A. (1953). "Ecological Regressions and Behavior of Individuals." *American Sociological Review* 18(6): 663-664.
- Duncan, O. D. and Davis, B. (1953). "An Alternative to Ecological Correlation." *American Sociological Review* 18(6): 665-666.
- Goodman, L. A. (1959). "Some Alternatives to Ecological Correlation." *American Journal of Sociology* 64(6): 610-625.
- Openshaw, S. (1984). *The Modifiable Areal Unit Problem*. Concepts and Techniques in Modern Geography 38. Norwich: Geo Books.
- Freedman, D. A., Klein, S. P., Sacks, J., Smyth, C. A. and Everett, C. G. (1991). "Ecological Regression and Voting Rights." *Evaluation Review* 15(6): 673-711.
- King, G. (1997). *A Solution to the Ecological Inference Problem: Reconstructing Individual Behavior from Aggregate Data*. Princeton: Princeton University Press.
- *Thornburg v. Gingles*, 478 U.S. 30 (1986).